

Submission LINALG1-PS1-SUB06

Question LINALG1-PS1-Q01

notation changes halfway:

1. $\det \approx 6(5) - 1(4) \approx 26$
2. Eliminate y , yielding $x \approx 1$; substitute $\Rightarrow y \approx 2$
3. checks: $6(1) + 1(2) \approx 8$, $4(1) + 5(2) \approx 14$

marks, subscripts, and units are mixed up

result written as: $x=1$, $y=2$ (please interpret)

Question LINALG1-PS1-Q02

notation changes halfway:

1. $\det(A) \approx 2 \cdot 3 - 2 \cdot 2 \approx 2$
2. $\det(A) \neq 0$, so A is invertible.

marks, subscripts, and units are mixed up

result written as: $\det=2$; invertible (please interpret)

Question LINALG1-PS1-Q03

notation changes halfway:

1. $T(1, -1) \approx (1 - 2, 3 + 1) \approx (-1, 4)$
2. Matrix $[[1, 2], [3, -1]]$
3. Both coordinate rules have no constant term, so T is linear.

marks, subscripts, and units are mixed up

result written as: $T(1, -1) = (-1, 4)$; matrix $[[1, 2], [3, -1]]$; linear (please interpret)

Question LINALG1-PS1-Q04

notation changes halfway:

1. Set $c_1(1, 0) + c_2(0, 1) \approx (0, 0)$.
2. Coordinates give $c_1 \approx 0$ and $c_2 \approx 0$ only, so the set is independent.

marks, subscripts, and units are mixed up

result written as: only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent (please interpret)

Question LINALG1-PS1-Q05

notation changes halfway:

1. $\text{Null}(A)$ is the set of vectors sent to the zero vector.
2. It is the solution space of $Ax \approx 0$ and is a subspace.

marks, subscripts, and units are mixed up

result written as: the vectors mapped to zero; exactly the solution space of $Ax=0$ (please interpret)

Question LINALG1-PS1-Q06

notation changes halfway:

1. $\det \approx 5(3) - 2(3) \approx 9$
2. Eliminate y , yielding $x \approx 1$; substitute $\Rightarrow y \approx 2$
3. checks: $5(1) + 2(2) \approx 9$, $3(1) + 3(2) \approx 9$

marks, subscripts, and units are mixed up
result written as: $x=1$, $y=2$ (please interpret)